Fisika 2

No.: Tugas Fisika Date.:

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Kelas : C1.25

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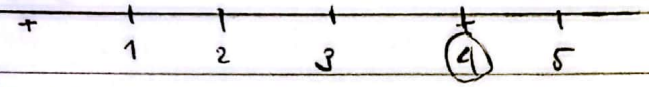
1. tentukan medan listrik di titik (0,3) akibat dua muatan berikut

$q_1 = +8nC$ $q_2 = +12nC$



$E = \frac{k|q|}{r^2}$

$k = 9 \times 10^9 \ Nm^2/C^2$

$q_1 = 8 \quad q_2 = 12$

r = Jarak

$q_1 = 8 \ (1,0)$

$r = \sqrt{(0-1)^2 + (3-0)^2} = \sqrt{10}$

$E_1 = \frac{(9 \times 10^9)(8 \times 10^{-9})}{\sqrt{10^2}} = 7.2 \ N/C$

2. medan listrik dari q_2 (E_2) jarak r_2 dari q_2 (4,0) ke titik (0,3)

$r = \sqrt{(0-4)^2 + (3-0)^2} = \sqrt{(-4)^2 + 3^2} = \sqrt{16+9} = \sqrt{25} = 5$

besar medan listrik E_2 :

$E_2 = \frac{k \cdot q_2}{r_2^2} = \frac{(9 \times 10^9) \cdot (12 \times 10^{-9})}{5^2} = \frac{108}{25} = 4,32 \ N/C$

total : $E_x = -2.28.3,456 = -5736 \ N/C$

$E_y = 6,83 + 2,592 = 9,422 \ N/C$

$E = \sqrt{E_x^2 + E_y^2} \simeq 11,03 \ N/C$

KIKY

Dipindai dengan CamScanner